CERTIFICATE OF PARTICIPATION

This certificate is proudly presented to

Nour Mahjoub

for successfully completing the training

"Altium Designer Pro - From Design to Manufacturing"

Organized and delivered by Thameur CHEBBI

Training Details

- **Date: 20/09/2025 to 25/10/2025**
- **Format:** Online
- **Duration:** 26 hours
- **Instructor:** Thameur CHEBBI

Key Competencies Acquired:

1. Mastered **advanced Altium Designer** workflows: **design rules** setup, **impedance control**, **stackup** configuration.
2. Designed robust **2-layer** and **4-layer** PCBs with optimized **return paths**, **controlled routing**.
3. Created **IPC-compliant symbols** and **footprints** (**IPC-7351 / IPC-2221/2222**), including validated **padstacks**, **courtyards**, and **3D models**.
4. Applied **high-speed** & **mixed-signal** PCB design principles: **differential routing**, **length tuning**, **impedance control**, and **EMI/EMC**-aware layout.
5. Integrated **SPI**, **I²C**, **UART**, and **USB interfaces** with proper routing constraints, and **signal integrity** considerations.
6. Performed **circuit simulation** and validation in **LTspice**, filtering, and design verification**.**
7. Generated complete fabrication & assembly files: **Gerber/ODB++, IPC-2581**, **pick-and-place**, **stackup sheets**, and **assembly documentation**.
8. **Developed**, **tested**, and **debugged ATmega328P**-based PCB using **ICSP** and **bootloader** workflows, applying structured **troubleshooting** practices.
9. **Developed**, **tested**, and **debugged STM32F103**-based PCB using **SWD** and **bootloader** workflows, applying structured hardware **troubleshooting** practices

Thameur Chebbi

Thameur CHEBBI
Senior Embedded Hardware & PCB Design Engineer
Advanced PCB Design Instructor